Full support and positive density in the zero-temperature SK model

A reconstruction of OpenAI's coercive-curvature argument · Revised 24 October 2026

Abstract. Every admissible integrable minimiser of the zero-temperature Parisi functional for the pure, zero-field Sherrington–Kirkpatrick model has full relative Stieltjes support on $[0,1)$ and a smooth density that is strictly positive on $(0,1)$. We reconstruct OpenAI's proof of these conclusions. Minimality gives contact conditions at the endpoints of a proposed support gap. An integrated coercive estimate makes the derivative of a diffusion moment strictly convex on each positive constant interval of the order parameter. The resulting area deficit contradicts contact; separate arguments handle zero coefficients and the two endpoints. A fixed-time approximation by shrinking constant intervals reuses the same estimate to prove strict positivity of the density. A martingale calculation recovers the scalar ground-state value. Chen already proved full support and smooth absolute continuity. Our contribution is a compact account of the two uses of the quantitative estimate and their distinct analytic requirements.

1. The order parameter and the result

We study how a minimising order parameter distributes mass across overlap scales. We use the pure SK covariance $\xi(t) = t^2/2$ and zero external field. Let $\mathcal{U}$ be the nonnegative, nondecreasing, right-continuous functions $\gamma : [0,1) \to [0,\infty)$ with $\int_0^1 \gamma(t)\,dt < \infty$. Define their Stieltjes measures by

$$\mu_\gamma([0,t]) = \gamma(t), \qquad 0 \le t < 1.$$

In particular, an atom of mass $\gamma(0)$ is initially allowed at zero. The measure is locally finite, need not have finite total mass, and is not normalised as a probability measure. Support is relative to $[0,1)$.

Let $\Phi = \Phi_\gamma$ solve the backward Parisi equation

$$\partial_t \Phi = -\frac{1}{2}(\Phi_{xx} + \gamma(t)\Phi_x^2), \qquad \Phi(1,x) = |x|,$$

and set

$$\mathcal{P}(\gamma) = \Phi_\gamma(0,0) - \frac{1}{2}\int_0^1 t\gamma(t)\,dt.$$

Write $u = \Phi_x$ and $a = \Phi_{xx}$. The associated diffusion is

$$dX_t = \gamma(t)u(t,X_t)\,dt + dB_t, \qquad X_0 = 0,$$

where B is standard Brownian motion. For an integrable coefficient, the value function is defined by

$$\Phi_\gamma(t,x) = \sup_{|\alpha|\le 1} \mathbb{E}\left[\left|x + B_1 - B_t + \int_t^1 \gamma(s)\alpha_s\,ds\right| - \frac{1}{2}\int_t^1 \gamma(s)\alpha_s^2\,ds\right],$$

where the controls are progressively measurable for the Brownian increments after t. It is equivalently obtained by monotone-coefficient approximation. The control and heat-equation methods have antecedents in Jagannath and Tobasco (2016) at positive temperature and Chen, Handschy and Lerman (2018), Proposition 2 and Theorem 5 at zero temperature. We use the estimates of

OpenAI (2026a), Lemmas 2.1–2.2: Φ is even, convex and one-Lipschitz in space, all its spatial derivatives are bounded and continuous on $[0,T]\times\mathbb{R}$ for $T<1$, and the PDE holds almost everywhere in time. In particular $|u|\le 1$, and

$$0 \le \Phi_\gamma(t,x) - |x| \le \sqrt{2(1-t)/\pi} + \frac{1}{2}\int_t^1 \gamma(s)\,ds. \tag{T}$$

Existence and uniqueness of the diffusion, these regularity estimates, and the variation formula below are analytic inputs from that source.

For independent standard Gaussian J_{ij} and $\sigma\in\{-1,1\}^n$, put $H_n(\sigma) = n^{-1/2}\sum_{i<j} J_{ij}\sigma_i\sigma_j$, the model introduced by Sherrington and Kirkpatrick (1975). The Parisi formula and attainment give

$$P_* := \lim_{n\to\infty}\frac{1}{n}\mathbb{E}\max_\sigma H_n(\sigma) = \min_{\gamma\in\mathcal{U}}\mathcal{P}(\gamma)$$

Auffinger and Chen (2017), Theorem 1 and proof of Lemma 3. Here $\mathrm{Cov}(H_n(\sigma),H_n(\tau)) = nR^2/2 - 1/2$ for overlap $R = n^{-1}\sum_i \sigma_i\tau_i$; a common centred Gaussian of variance $1/2$ restores $n\xi(R)$ without changing the expected maximum.

Theorem 1 (full support; OpenAI (2026a), Theorem 1.1). If $\gamma\in\mathcal{U}$ minimises $\mathcal{P}$, then

$$\operatorname{supp}\mu_\gamma = [0,1), \qquad \gamma(t_2) > \gamma(t_1) \quad (0\le t_1<t_2<1).$$

For every associated diffusion,

$$\mathbb{E}[u(t,X_t)^2] = t, \qquad \mathbb{E}[a(t,X_t)^2] = 1 \quad (0\le t<1).$$

The theorem assumes a minimiser; its existence comes from the separate attainment result. Integrability permits growth of γ near one. Zero field supplies the symmetry $u(0,0)=0$, and the pure-SK value $\xi''=1$ enters the curvature calculation. No extension to nonzero field or a general mixed-spin covariance is asserted.

Infinite support alone leaves room for gaps: the measure $\sum_{k\ge1} 2^{-k}\delta_{1-2^{-k}}$ is an example. Auffinger, Chen and Zeng (2020), Theorem 1 proved infinite support; Chen (2026), Theorem 1.3 proved full relative support and smooth absolute continuity. Our route is OpenAI's quantitative curvature argument. Its second use will prove the separate positive-density Corollary 3 in §4. The fixed-revision Lean implementation and formal challenge specify the narrower support-and-moment interface; their Comparator configuration selects an intended comparison, with the actual evidence described in §6.

2. Contact and the contradiction inside a gap

Fix a minimiser and define

$$q(t) = \mathbb{E}[u(t,X_t)^2], \qquad G(s) = \int_s^1 (q(t)-t)\,dt.$$

Itô's formula gives $du(t,X_t) = a(t,X_t)\,dB_t$, hence $q(0)=0$ and $q'(t) = \mathbb{E}[a(t,X_t)^2]$. The compact-time regularity makes q' continuous, including across jumps of γ. The one-sided first variation is

$$D_+\mathcal{P}(\gamma)[\eta] = \frac{1}{2}\int_0^1 \eta(t)(q(t)-t)\,dt$$

for every integrable direction η such that $\gamma + \varepsilon\eta \in \mathcal{U}$ for sufficiently small $\varepsilon \geq 0$ OpenAI (2026a), Lemma 6.1. The earlier zero-temperature variation and support-point identities are in Chen, Handschy and Lerman (2018), Propositions 3–4.

Adding an atom at s means taking $\eta = \mathbf{1}_{[s,1)}$, so minimality gives $G(s) \geq 0$. To obtain equality on the support, take a compact $K \subset [0,1)$ and let ν be the finite restriction of μ_γ to K. Removing a fraction of this mass is feasible and gives

$$0 \leq D_+\mathcal{P}(\gamma)[-\nu([0,\cdot])] = -\frac{1}{2}\int_K G(s)\, d\mu_\gamma(s).$$

Thus $G = 0$ almost everywhere for μ_γ, and by continuity on $S = \operatorname{supp}\mu_\gamma$. This argument removes diffuse mass as well as atoms; no density assumption or probability constraint is used.

At a support point $s > 0$, G has a minimum. Since $G' = s - q(s)$ and $G'' = 1 - q'$, we obtain

$$q(s) = s, \qquad q'(s) \leq 1. \tag{1}$$

If $0 \in S$, then $G(0) = q(0) = 0$ and $G(h) = \frac{1}{2}(1 - q'(0))h^2 + o(h^2)$, so the right derivative also satisfies $q'(0) \leq 1$. These are the contact conditions of OpenAI (2026a), Lemma 6.2.

A support gap forces γ to be constant there. The analytic estimate that rules it out does not require minimality.

Lemma 2 (constant-interval curvature; OpenAI (2026a), Proposition 5.1). For any $\gamma \in \mathcal{U}$, define q using its associated diffusion as above. If $\gamma = c$ on an open interval $I \subset (0,1)$, then

$$q'''(t) \geq \frac{c^2}{1000}\mathbb{E}[a(t, X_t)^4] > 0 \quad (t \in I,\ c > 0). \tag{2}$$

Thus q' is strictly convex on a positive constant interval. If $c = 0$, then $q''(t) = \mathbb{E}[\Phi_{xxx}(t, X_t)^2] > 0$, so q' is strictly increasing there.

We first use this estimate, then explain its proof. Suppose (ℓ, r) is a gap with both endpoints in S and $r < 1$. On a positive plateau, strict convexity and (1) give

$$q'(t) < \frac{r - t}{r - \ell}q'(\ell) + \frac{t - \ell}{r - \ell}q'(r) \leq 1 \quad (\ell < t < r).$$

Consequently $q(r) - q(\ell) < r - \ell$, contradicting contact at both endpoints. The geometric obstruction is an area deficit under q'. On a zero plateau, strict increase instead gives $q'(t) < q'(r) \leq 1$ and the same contradiction. An initial gap $[0, r)$ with $r \in S$, $r < 1$ and $0 \notin S$ has $\gamma = 0$, including no atom at zero; use $q(0) = 0$ and contact at r to reach that contradiction without assuming contact at zero.

The point 1 lies outside the support domain, so a terminal gap needs a different argument. On a constant interval ending at one, OpenAI (2026a), Lemma 6.3 proves

$$1 - q(t) \geq C\sqrt{1 - t}$$

for some $C > 0$ and all sufficiently late t. In a window of width $\sqrt{1-t}$ around the terminal cusp, u stays uniformly away from ± 1 and the diffusion density has a uniform positive lower bound. Integrating $1 - u^2$ over that window gives the estimate. Hence $q(t) - t \leq (1 - t) - C\sqrt{1 - t} < 0$

near one, contradicting $G \geq 0$. This also excludes empty support. Terminal-plateau exclusion was already recorded by Auffinger, Chen and Zeng (2020), Remark 7.

These cases exhaust the components of the complement of the relatively closed set S. Thus $S = [0, 1)$, (1) gives $q(t) = t$ everywhere, and differentiation gives $\mathbb{E}a^2 = 1$, including the right-hand value at zero. Positive mass on every open interval gives $\gamma(t_2) - \gamma(t_1) = \mu_\gamma((t_1, t_2]) > 0$. This proves Theorem 1 conditional on Lemma 2 and the cited analytic inputs.

3. What makes the curvature positive

On a positive plateau put $r = c\Phi_x$, $v = r_x$, $w = r_{xx}$ and $z = r_{xxx}$. Along X, their drift and diffusion pairs are $(0, v)$, $(-v^2, w)$ and $(-3vw, z)$ for r, v, w, respectively. Starting from $c^2 q' = \mathbb{E}v^2$, Itô's formula yields

$$c^2 q'' = \mathbb{E}[w^2 - 2v^3], \qquad c^2 q''' = \mathbb{E}[z^2 - 12vw^2 + 6v^4]. \tag{3}$$

The drift of w^2 is $z^2 - 6vw^2$; that of v^3 is $3vw^2 - 3v^4$. The negative mixed term is the difficulty: convexity of Φ does not give a sign for this expression.

First take a positive nondecreasing finite-step coefficient and smooth the terminal value to $M^{-1} \log(2 \cosh Mx)$, with $M \geq \|\gamma_{\text{step}}\|_\infty$; along approximation sequences we also take $M \to \infty$. On a step of height c, the diffusion density factors as $f = e^{c\Phi} g$, where g solves the forward heat equation. For the score $s = -\partial_x \log g$, the source propagates inequalities jointly with those for r: on $x > 0$, r and s are increasing and concave, while $w/(2r) + v \leq 0$ is nondecreasing and s/r is nondecreasing. The relevant ratios extend through zero. These are analytic shape inequalities, proved in §3 of OpenAI (2026a). The Cole–Hopf and transformed-density methods have close antecedents in Lopatto (2026), Propositions 3.6, 7.1 and 9.1 and Chen (2026), §§3–4 and Appendix C; Chen's appendix distinguishes the earlier shared estimates from the parallel crossing proofs.

The rational certificate of OpenAI (2026a), Lemma 4.1 and Proposition 4.2 adds a polynomial derivative correction. For its polynomial F, the associated expression $\mathcal{W}(F)$ satisfies

$$f\mathcal{W}(F) = \partial_x(rfF).$$

After this correction the shape inequalities make the remaining weighted squares and polynomial terms nonnegative, leaving at least $10^{-3}v^4$. The source's tail bounds remove the derivative upon integration. Therefore

$$\mathbb{E}_f[z^2 - 12vw^2 + 6v^4] \geq \frac{1}{1000}\mathbb{E}_f[v^4]. \tag{4}$$

This is an integrated inequality for the specified profiles and density; the polynomial on the left has no such sign for arbitrary v, w, z.

For a general integrable coefficient, positive monotone finite-step approximants retain the plateau height c near a fixed interior time. Spatial derivatives through order four converge uniformly on compact time strips, and the coupled diffusions converge. Only v, w, z remain in (4), so convergence of the auxiliary scores is unnecessary. Strict positivity of Φ_{xx} gives (2). On a zero plateau, heat evolution gives the separate formula for q'' in Lemma 2. It is strict at positive times because the diffusion has a positive density and Φ_{xxx} cannot vanish identically: a quadratic spatial profile would contradict the one-Lipschitz bound and $|x| + O(1)$ growth.

Chen's Proposition 4.2 gives the crossing implication $q''(t) = 0 \Rightarrow q'''(t) > 0$ on a hypothetical positive gap of the minimiser Chen (2026). Estimate (2) is quantitative and holds throughout each

positive plateau for any admissible coefficient. The complete sign, certificate and approximation proofs are in OpenAI (2026a), §§2–5; the long reasoning record, §4, "Strict curvature on a constant interval", explains their assembly.

4. The same estimate makes the density positive

Full support does not by itself imply a positive density at every point. For example, the smooth increasing function $\gamma(t) = \int_0^t (s - \frac{1}{2})^2 \, ds$ has full Stieltjes support but $\gamma'(\frac{1}{2}) = 0$. The source's coercive estimate rules out this behaviour for a minimiser.

Corollary 3 (smooth positive density; OpenAI (2026a), Corollary 6.4). Under the hypotheses of Theorem 1, $\gamma(0) = 0$ and $\gamma \in C^\infty([0,1))$, with derivatives at zero understood from the right. Moreover, $\mu_\gamma(dt) = \gamma'(t)\, dt$, and

$$2\gamma'(t)\mathbb{E}[a(t, X_t)^3] \geq \frac{\gamma(t)^2}{1000}\mathbb{E}[a(t, X_t)^4] > 0 \quad (0 < t < 1). \tag{5}$$

In particular the density is strictly positive on $(0, 1)$; only nonnegativity at zero is asserted. This is a separate manuscript result, not a conclusion of the selected full-support formal interface.

For the proof, write $b = \Phi_{xxx}$ and $d = \Phi_{xxxx}$, evaluating spatial factors along X below. Since $da = -\gamma a^2\, dt + b\, dB$ and $\mathbb{E}a^2 = 1$, Itô's formula gives

$$\gamma(t) = \frac{\mathbb{E}b(t, X_t)^2}{2\mathbb{E}a(t, X_t)^3}. \tag{6}$$

Initially the identity holds almost everywhere. Convexity gives $a \geq 0$ and $\mathbb{E}a^3 \geq (\mathbb{E}a^2)^{3/2} = 1$. Compact-strip continuity of the moments and right continuity of γ extend (6) to every time, proving continuity of γ. Evenness gives $b(0, 0) = 0$, so $\gamma(0) = 0$.

To bootstrap, consider all polynomial moments of finitely many spatial derivatives at once. Itô's formula writes the derivative of each as another polynomial moment plus γ times a polynomial moment. Their continuity makes them C^1; (6) then makes γ C^1. Repeating gives smoothness without assuming derivatives of γ in advance. This smoothness argument is in Chen (2026), Proposition 4.6 and Appendix B.6; the polynomial-moment method is developed in Auffinger and Chen (2015), Lemma 2 and the proof of Theorem 2(ii).

Differentiating $\mathbb{E}b^2 = 2\gamma\mathbb{E}a^3$ now gives

$$2\gamma'\mathbb{E}a^3 = \mathbb{E}[d^2 - 12\gamma ab^2 + 6\gamma^2 a^4]. \tag{7}$$

Fix $0 < t < 1$ and put $c = \gamma(t) > 0$. Flatten γ to c on a shrinking interval about t, preserving monotonicity, and approximate by positive finite-step coefficients with that plateau. These coefficients converge in L^1. With the terminal smoothings of §3, apply (4) at the fixed time, with $(v, w, z) = (ca, cb, cd)$. The derivative and diffusion convergence on a fixed compact strip does not depend on the shrinking plateau widths, so the inequality passes to the original coefficient. Equations (4) and (7) yield (5), with strictness from $\mathbb{E}a^4 \geq 1$. The long record, §§5.1–5.2, retains the moment bootstrap and the fixed-time approximation proving strict positivity of the density.

5. The scalar value and the further circuit problem

There is also a direct value consequence, which needs full support but not the density corollary. The gradient-martingale and normalised Euler methods occur in Montanari (2025), §§2–3 of the

cited preprint and El Alaoui, Montanari and Sellke (2021), §§5–6 of the cited preprint. For a minimiser and its associated diffusion, the bounded martingale $M_t = u(t, X_t) = \int_0^t a(s, X_s)\, dB_s$ satisfies $\mathbb{E}M_t^2 = t$. It therefore converges almost surely and in L^2 to a variable U with $|U| \le 1$ and $\mathbb{E}U^2 = 1$, so $U^2 = 1$ almost surely. The integrable drift bound $|\gamma u| \le \gamma$ gives $X_t \to X_1$ almost surely and in L^2.

The terminal estimate (T) and convexity imply $u(t, x_t) \to \operatorname{sgn}(x)$ whenever $x_t \to x \neq 0$: trap the derivative between difference quotients on a small interval that stays on one side of zero. Thus $UX_1 = |X_1|$, including when $X_1 = 0$. Itô's formula gives

$$\mathbb{E}\Phi_\gamma(t, X_t) = \Phi_\gamma(0, 0) + \frac{1}{2}\int_0^t \gamma(s) s\, ds.$$

Using (T) to let $t \uparrow 1$ and subtracting the Parisi penalty yields $\mathcal{P}(\gamma) = \mathbb{E}|X_1| - \int_0^1 t\gamma(t)\, dt$. Multiply $X_1 = B_1 + \int_0^1 \gamma(t) M_t\, dt$ by U and use $\mathbb{E}[UM_t] = \mathbb{E}M_t^2 = t$. Boundedness and $\gamma \in L^1$ justify Fubini, so the drift cancels the penalty. Finally $\mathbb{E}\int_0^1 a(t, X_t)^2\, dt = 1$ extends the stochastic integral in L^2 to time one; its cross-isometry with B_1 gives

$$P_* = \mathbb{E}[UB_1] = \int_0^1 \mathbb{E}[a(t, X_t)]\, dt. \tag{8}$$

This is the first curvature moment, distinct from $\mathbb{E}a^2 = 1$; its integrability follows by Cauchy–Schwarz. The equality with P_* uses the external Parisi formula. This proves the scalar identity of OpenAI (2026a), Corollary 7.1; the long record, §6, "Recovering the value from a martingale", retains the full argument. The scalar-value implementation, formal challenge and comparison configuration are separate from those for full support.

The companion finite-Gaussian approximation first refines a mesh on $[0, T]$ at fixed $0 < T < 1$, then sends $T \uparrow 1$. Its scalar Gaussian correlations are not themselves circuit energies OpenAI (2026a), Proposition 7.2; see the long record, §7, "A finite Gaussian spin and its adjacent correlations". The order also gives a finite accuracy choice: since $a \ge 0$ and $\mathbb{E}a^2 = 1$,

$$0 \le P_* - \int_0^T \mathbb{E}[a(t, X_t)]\, dt \le 1 - T.$$

For $0 < \varepsilon < 1$, choose $T = 1 - \varepsilon/2$, then a finite mesh whose coefficient sum differs from the fixed-T integral by less than $\varepsilon/2$. This provides a finite scalar approximation within ε, without specifying a mesh rate or a coupled cutoff schedule.

The quantum approximate optimisation algorithm of Farhi, Goldstone and Gutmann (2014) alternates cost-Hamiltonian evolution and transverse-field mixing from the uniform superposition. At depth p, let $v_p(\theta, \beta)$ be its limiting disorder-averaged energy per spin, with deterministic cost and mixer angles held fixed as $n \to \infty$. This limit and fixed-parameter concentration are due to Farhi, Goldstone, Gutmann and Zhou (2022). Their minimum-energy convention becomes the maximum-energy convention here by symmetry of the disorder and reversal of the cost angles. Put

$$Q_p = \sup_{\theta, \beta \in \mathbb{R}^p} v_p(\theta, \beta).$$

The further theorem of OpenAI (2026b), Theorem 1.1 asserts $\lim_{p\to\infty} Q_p = P_*$, the eventual-optimality conclusion conjectured by Basso et al. (2022), §7, equation (7.2) of the full preprint.

Equivalently, for every $\varepsilon > 0$ there are a finite depth and deterministic angles, independent of both n and the disorder, whose limiting expected energy is at least $P_* - \varepsilon$. The depth and angles may depend on ε. The system-size limit is taken at that fixed circuit before improving the accuracy; the theorem provides no quantitative depth bound, finite-size rate or efficient angle-selection procedure.

The concentration result also turns the expected-energy conclusion into a sampling statement. Choose the word to attain $P_* - \varepsilon/2$. Its first and second energy moments converge to v_p and v_p^2, so a measured energy per spin is at least $P_* - \varepsilon$ with probability tending to one jointly over disorder and quantum measurement. The long record, §9.8, gives the mean-square and Markov argument; this consequence still supplies no convergence rate or every-instance guarantee.

The scalar Gaussian correlations above do not yet provide such a circuit. OpenAI constructs a rooted representation of the Gaussian calculation, synthesises it with quantum controls, removes temporary longitudinal seed controls, and transfers one fixed finite circuit to SK. It also gives a positive-temperature construction with different classical input. The long record, §9, "From the scalar formula to one ordinary QAOA word", reconstructs these additional arguments and their antecedents; the originals are OpenAI (2026b), §§2–6 and Appendix A.

6. Source and verification scope

This AI-assisted draft reconstructs existing results credited to OpenAI and the cited authors. The main mathematical work is the integrated coercivity estimate: variation turns it into full support, and saturated moments turn it into positive density. Neither this reconstruction nor its finite algebra check establishes a new theorem.

The selected formal support statement includes measure and diffusion existence, support, strict increase and both moment identities for every associated diffusion, conditional on a minimiser. The separate scalar statement assumes a minimiser and an associated diffusion for its scalar and finite-Gaussian clauses; its ground-state/Parisi equality assumes only the minimiser. Neither statement includes Corollary 3 or QAOA convergence.

Source-integrity checks passed. After an earlier unavailable Comparator startup, a later recorded attempt built and exported the admitted full-support challenge, then failed to build the implementing solution because an imported compiled object was missing. No successful corrected replay or completed scalar-target replay is recorded. Challenge export, tool-fixture success and the exact polynomial calculation establish no research-target acceptance, transitive axiom audit or independent human review. Exact versions, the partial attempt and remaining obligations are retained in the companion source review.

References

- OpenAI (2026a). *Full support of the zero-temperature Sherrington–Kirkpatrick order parameter*. 27 September 2026. Manuscript, fixed revision. Theorem 1.1; §§2–6; Corollaries 6.4 and 7.1; Proposition 7.2.
- OpenAI (2026b). *QAOA attains the SK ground-state energy in the thermodynamic-first limit*. 25 September 2026. Manuscript, fixed revision. Theorem 1.1; §§2–6 and Appendix A.
- Auffinger, A. and Chen, W.-K. (2017). "Parisi formula for the ground state energy in the mixed p-spin model." *Annals of Probability* **45**, 4617–4631. Preprint v2: Theorem 1 and proof of Lemma 3.

- Auffinger, A., Chen, W.-K. and Zeng, Q. (2020). "The SK model is infinite step replica symmetry breaking at zero temperature." *Communications on Pure and Applied Mathematics* **73**, 921–943. Preprint v2: Theorem 1 and Remark 7.
- Chen, H.-B. (2026). *FRSB in the SK spin glass: convergence to full-interval support at zero temperature*. Version 1, 20 July 2026. Theorem 1.3; Propositions 4.2 and 4.6; Appendices B.6 and C.
- Lopatto, P. (2026). *Full replica symmetry breaking in the Sherrington–Kirkpatrick model*. arXiv version 3, submitted 16 July 2026. Propositions 3.6, 7.1 and 9.1; related Cole–Hopf, density and crossing arguments.
- Sherrington, D. and Kirkpatrick, S. (1975). "Solvable Model of a Spin-Glass." *Physical Review Letters* **35**, 1792–1796. Original model.
- Jagannath, A. and Tobasco, I. (2016). "A dynamic programming approach to the Parisi functional." *Proceedings of the AMS* **144**, 3135–3150. Preprint v4. Positive-temperature control methods.
- Chen, W.-K., Handschy, M. and Lerman, G. (2018). "On the energy landscape of the mixed even p-spin model." *Probability Theory and Related Fields* **171**, 53–95. Preprint v4: Proposition 2, Theorem 5, Lemma 3 and Propositions 3–4.
- Auffinger, A. and Chen, W.-K. (2015). "On properties of Parisi measures." *Probability Theory and Related Fields* **161**, 817–850. Preprint v1: Lemma 2 and proof of Theorem 2(ii).
- Montanari, A. (2025). "Optimization of the Sherrington–Kirkpatrick Hamiltonian." *SIAM Journal on Computing* **54**, FOCS19-1–FOCS19-38. Preprint v2, 2019: §§2–3.
- El Alaoui, A., Montanari, A. and Sellke, M. (2021). "Optimization of mean-field spin glasses." *Annals of Probability* **49**, 2922–2960. Preprint v1: §§5–6.
- Farhi, E., Goldstone, J. and Gutmann, S. (2014). *A Quantum Approximate Optimization Algorithm*. Preprint v1. Algorithm definition; §VI concerns the fixed-instance depth limit.
- Farhi, E., Goldstone, J., Gutmann, S. and Zhou, L. (2022). "The Quantum Approximate Optimization Algorithm and the Sherrington–Kirkpatrick Model at Infinite Size." *Quantum* **6**, 759. Preprint v4.
- Basso, J., Farhi, E., Marwaha, K., Villalonga, B. and Zhou, L. (2022). "The Quantum Approximate Optimization Algorithm at High Depth for MaxCut on Large-Girth Regular Graphs and the Sherrington–Kirkpatrick Model." *TQC 2022*, 7:1–7:21. Full preprint v3: transfer theorem and §7, equation (7.2); the full preprint has separate numbering.